test input, full runs — small-dm

L1D 64 sets x 1, L2 512 sets x 1 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

35 of 35 runs completed, 0 running, 0 pending; **0 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	1/1	21.32%	51.31%	55.47%	21.42%	51.49%	55.80%
parest	1/1	0.22%	0.56%	1.65%	0.81%	1.48%	12.55%
deepsjeng	1/1	2.78%	9.33%	15.84%	7.31%	22.43%	46.66%
leela	1/1	0.36%	1.25%	1.66%	6.11%	15.91%	44.30%
namd	1/1	3.13%	7.91%	17.27%	3.14%	7.94%	17.37%
lbm	1/1	9.67%	27.79%	35.34%	9.67%	27.79%	35.34%
nab	1/1	7.90%	26.29%	39.86%	8.63%	27.88%	47.43%
xz	1/1	0.68%	2.35%	2.75%	3.30%	9.53%	17.13%
xz_4m_l1	1/1	0.59%	2.08%	2.43%	2.95%	8.51%	15.84%
xz_4m_l2	1/1	0.54%	1.91%	2.32%	2.73%	7.91%	15.62%
xz_4m_l3e	1/1	2.51%	9.02%	25.13%	3.76%	12.59%	42.10%
xz_4m_l4	1/1	2.12%	6.97%	14.43%	4.34%	13.23%	32.30%
xz_4m_l4e	1/1	2.28%	8.15%	25.17%	3.39%	11.30%	42.07%
xz_1m_l0	1/1	0.62%	2.50%	1.68%	3.69%	10.95%	17.49%
xz_1m_l1	1/1	0.58%	2.38%	1.48%	3.48%	10.36%	16.63%
xz_1m_l2	1/1	0.57%	2.35%	1.45%	3.39%	10.13%	16.34%
xz_1m_l3e	1/1	2.59%	9.35%	16.76%	4.60%	15.17%	33.91%
xz_1m_l4	1/1	2.01%	6.65%	9.30%	5.08%	15.11%	26.95%
xz_1m_l4e	1/1	2.43%	8.77%	16.84%	4.27%	14.07%	33.92%
imagick	1/1	0.02%	0.08%	0.05%	4.16%	11.90%	27.17%
blender	1/1	1.16%	2.72%	4.44%	9.37%	20.20%	39.50%
povray	1/1	0.27%	0.45%	0.93%	14.17%	22.21%	55.05%
omnetpp	1/1	0.09%	0.19%	0.36%	12.82%	28.51%	53.40%
xalancbmk	1/1	0.23%	0.53%	0.92%	8.96%	16.83%	45.24%
x264	1/1	6.24%	13.47%	39.69%	8.48%	18.06%	54.69%
gcc	1/1	0.84%	2.18%	3.58%	12.72%	34.38%	52.28%
mcf	1/1	1.58%	2.11%	16.47%	2.74%	3.87%	27.40%
perlbench_makerand	1/1	0.08%	0.00%	0.50%	18.63%	28.89%	59.00%
bwaves_1	1/1	3.55%	9.47%	18.87%	3.55%	9.47%	18.87%
bwaves_2	1/1	3.55%	9.47%	18.87%	3.55%	9.47%	18.87%
exchange2	1/1	5.03%	10.75%	12.58%	6.23%	13.14%	15.81%
fotonik3d	1/1	3.84%	10.81%	9.25%	3.87%	10.86%	9.40%
roms	1/1	4.79%	15.22%	9.67%	4.88%	15.37%	10.29%
wrf	1/1	0.91%	3.41%	2.49%	1.59%	4.94%	7.83%
cam4	1/1	2.11%	7.90%	7.04%	4.52%	14.20%	21.40%
pooled		5.17%	12.33%	27.44%	7.17%	16.47%	39.99%
mean		2.78%	7.88%	12.36%	6.35%	15.89%	31.03%

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	17.36%	6.42%	36.76%	5.51%	16.60%	48.80%	11.27%	8.80%
parest	done	15.36%	20.70%	16.29%	6.63%	4.09%	47.58%	6.04%	4.98%
deepsjeng	done	13.69%	6.73%	17.17%	9.45%	7.46%	19.74%	5.47%	19.70%
leela	done	8.96%	19.60%	9.62%	0.46%	6.77%	17.75%	0.11%	89.76%
namd	done	15.58%	12.11%	18.89%	2.49%	3.89%	37.33%	9.19%	19.63%
lbm	done	27.22%	1.63%	13.94%	5.46%	81.64%	75.06%	5.17%	6.13%
nab	done	10.79%	1.54%	16.64%	1.65%	2.00%	15.30%	4.26%	1.90%
xz	done	8.70%	1.33%	10.47%	0.57%	5.73%	28.28%	0.77%	4.42%
xz_4m_l1	done	10.08%	1.45%	11.65%	0.59%	7.40%	36.30%	0.67%	6.24%
xz_4m_l2	done	11.65%	1.49%	13.72%	0.61%	7.75%	42.38%	0.72%	6.54%
xz_4m_l3e	done	9.40%	3.08%	10.73%	2.60%	7.62%	25.69%	2.17%	2.46%
xz_4m_l4	done	10.81%	3.07%	12.82%	3.15%	7.68%	33.53%	2.39%	6.32%
xz_4m_l4e	done	9.17%	3.19%	10.25%	2.62%	7.73%	24.56%	2.26%	2.47%
xz_1m_l0	done	7.98%	2.02%	10.31%	0.93%	4.06%	29.63%	0.74%	4.37%
xz_1m_l1	done	8.79%	2.14%	11.01%	0.94%	5.05%	33.78%	0.69%	4.81%
xz_1m_l2	done	8.92%	2.15%	11.17%	0.94%	5.09%	34.53%	0.69%	4.80%
xz_1m_l3e	done	8.68%	2.69%	10.71%	3.22%	5.26%	23.89%	1.69%	2.08%
xz_1m_l4	done	8.85%	2.90%	11.22%	4.94%	5.07%	28.51%	1.76%	3.37%
xz_1m_l4e	done	8.88%	2.79%	10.80%	3.23%	5.33%	22.40%	1.64%	2.05%
imagick	done	7.43%	16.95%	7.67%	70.35%	6.85%	47.92%	58.94%	85.67%
blender	done	9.92%	2.96%	11.28%	7.02%	7.67%	31.78%	70.70%	1.91%
povray	done	13.66%	13.93%	16.10%	21.86%	7.48%	37.82%	37.71%	55.51%
omnetpp	done	15.05%	0.72%	19.86%	10.67%	6.10%	36.64%	7.35%	0.74%
xalancbmk	done	15.16%	13.19%	18.38%	18.14%	6.36%	41.74%	10.63%	10.75%
x264	done	3.98%	0.98%	4.71%	1.20%	4.58%	26.05%	4.69%	17.50%
gcc	done	10.14%	2.27%	13.17%	4.98%	5.94%	32.44%	2.38%	7.12%
mcf	done	28.22%	8.95%	31.53%	10.17%	10.66%	72.37%	8.73%	19.17%
perlbench_makerand	done	15.27%	0.00%	19.33%	1.44%	7.06%	0.34%	—	0.01%
bwaves_1	done	44.12%	5.35%	49.10%	1.37%	47.45%	27.18%	19.22%	2.30%
bwaves_2	done	44.12%	5.35%	49.10%	1.37%	47.45%	27.18%	19.22%	2.30%
exchange2	done	4.01%	3.42%	5.09%	3.67%	2.66%	1.04%	0.14%	0.49%
fotonik3d	done	22.43%	9.89%	29.57%	0.43%	9.39%	38.96%	11.39%	2.22%
roms	done	30.01%	7.11%	37.22%	3.38%	21.21%	38.90%	15.88%	7.82%
wrf	done	26.37%	13.69%	28.28%	9.13%	21.78%	26.50%	14.64%	14.89%
cam4	done	14.02%	5.43%	14.76%	4.41%	14.73%	44.30%	11.42%	26.82%
pooled		11.58%	3.60%	14.20%	2.60%	9.46%	35.05%	9.96%	12.41%
mean		14.99%	5.92%	17.69%	6.44%	12.10%	33.03%	10.32%	13.03%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save

column	formula	aciklama
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.